

SHIVAM KUMAR VERMA

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PROFESSIONAL SUMMARY

Highly skilled Backend Software Engineer with a strong foundation in Python and JavaScript/TypeScript, experienced in developing scalable, high-performance backend systems and evaluating AI model logic. Proven ability to solve complex coding problems, optimize system performance, and design robust architectures. Eager to apply expertise in backend development, algorithms, and data structures to train and evaluate AI models, ensuring correctness and performance for DataAnnotation.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, C++

Backend Frameworks: Node.js, Express.js, FastAPI

Databases & ORMs: PostgreSQL, Redis, ClickHouse, Prisma, Drizzle

AI & ML Concepts: LLM integration, Prompt engineering, Semantic search, Pinecone

DevOps & Cloud: Docker, AWS (EC2, S3, Lambda), Nginx, Linux, Git, CI/CD

Tools & Methodologies: Algorithms, Data Structures, Debugging, Competitive Programming

EXPERIENCE

Software Developer Intern at klimb.io

Nov 2025 – Current • Remote

- Redesigned platform access control by migrating from feature flags to a credit-based usage system using PostgreSQL and middleware guards, enabling monetization of 6+ AI features and cutting complexity by ~70%.
- Replaced per-tenant field duplication with a shared global template model by restructuring the database schema and backfilling data, achieving zero-downtime migration with complete data integrity across all customers.
- Eliminated expensive runtime aggregations by introducing precomputed, indexed data pipelines, slashing recruiter dashboard load time from 2–3s to under 500ms.

PROJECTS

CodeSM [React](#), [Node.js](#), [Express.js](#), [Redis](#), [BullMQ](#), [Docker](#), [PostgreSQL](#), [AWS EC2](#)

Personal Project

- Engineered an asynchronous code execution engine by implementing segregated BullMQ queues and lightweight Redis payload caching, enabling reliable high-throughput submissions with near-zero dropped jobs.
- Eliminated container spin-up overhead by switching to persistent Docker containers with tmpfs mounts, reducing per-test execution time from 300ms to 5ms (~98% faster).
- Hardened job processing by adding a Dead Letter Queue for retries and a 1-hour Redis cache layer, significantly reducing database load during high-traffic spikes.

Storix [Next.js](#), [TypeScript](#), [FastAPI](#), [Celery](#), [Redis](#), [PostgreSQL](#), [AWS S3](#)

Personal Project

- Decoupled PDF ingestion from processing by introducing distributed Celery workers and a Redis task queue, enabling non-blocking long-running jobs with no impact on API response times.
- Offloaded file transfer from the server by generating pre-signed S3 URLs for direct client uploads, eliminating backend bottlenecks and improving upload reliability.
- Implemented a Pub/Sub and WebSocket layer to push live task updates, providing users with instant feedback from upload through delivery.

Chatterly [Next.js](#), [Socket.IO](#), [Redis](#), [PostgreSQL](#), [Prisma](#), [NextAuth](#)

Personal Project

- Enabled multi-instance chat by wiring Redis Pub/Sub for cross-server event propagation, ensuring consistent room state across all nodes without sticky sessions.
- Isolated room traffic by subscribing each server to dedicated Redis channels, guaranteeing zero message drops under concurrent multi-user load across distributed instances.
- Protected private rooms by enforcing JWT verification and room-level checks at the Socket.IO middleware layer, fully preventing unauthorized access.

EDUCATION

B.Tech in Computer Science and Engineering	from Indian Institute of Information Technology	Nov 2022 – May 2026	• Nagpur, Maharashtra
CGPA: 7.5 Relevant Coursework: Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming			